

LOSO Experiment B_flex_accel_lstm

LSTM only · Flex + Accel (both hands), no Nothing

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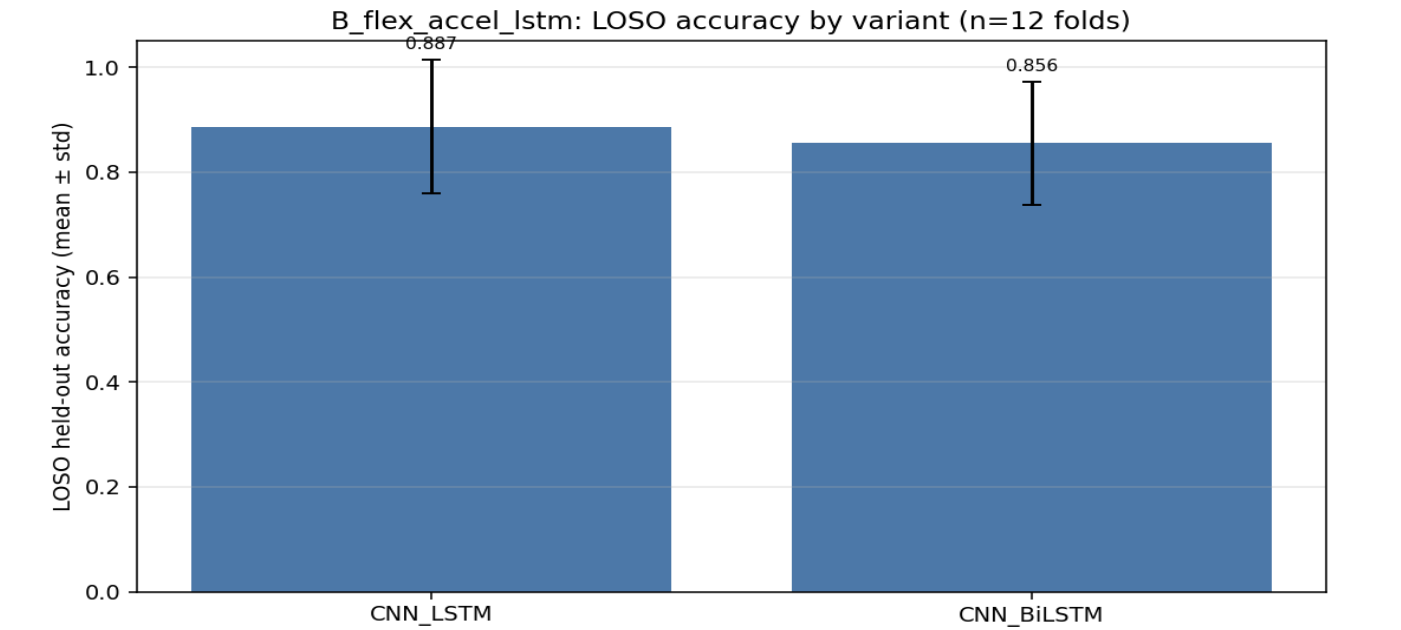
Experiment configuration

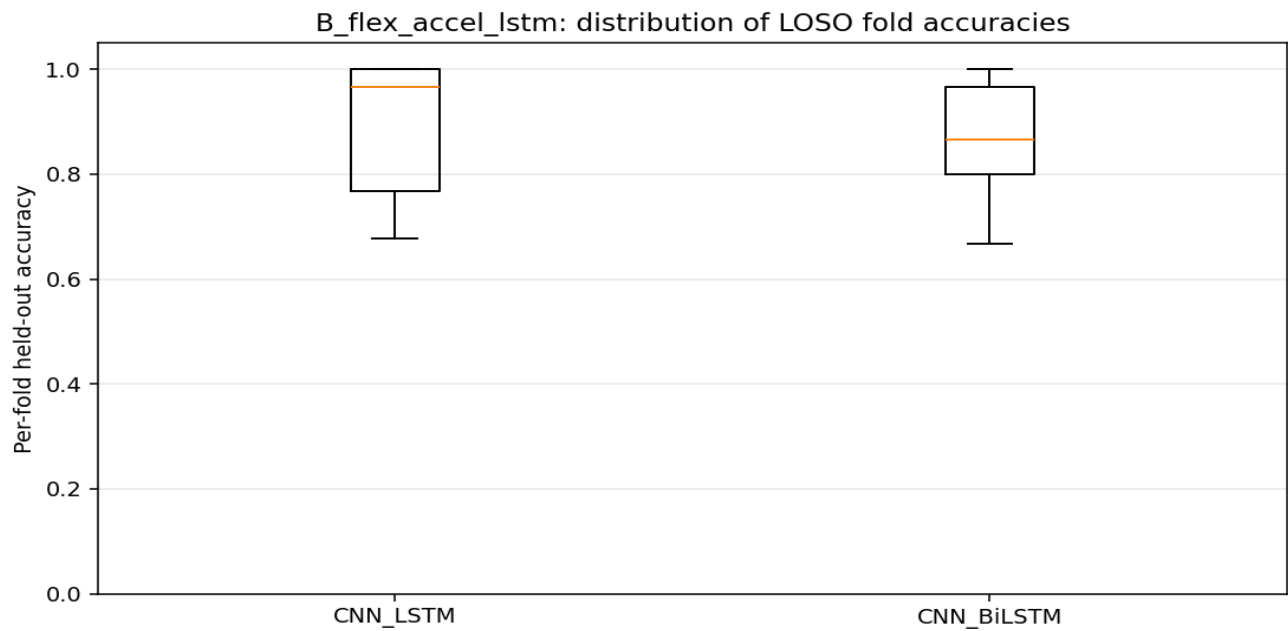
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, accel
Hands	left, right
Sensor channels	98
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

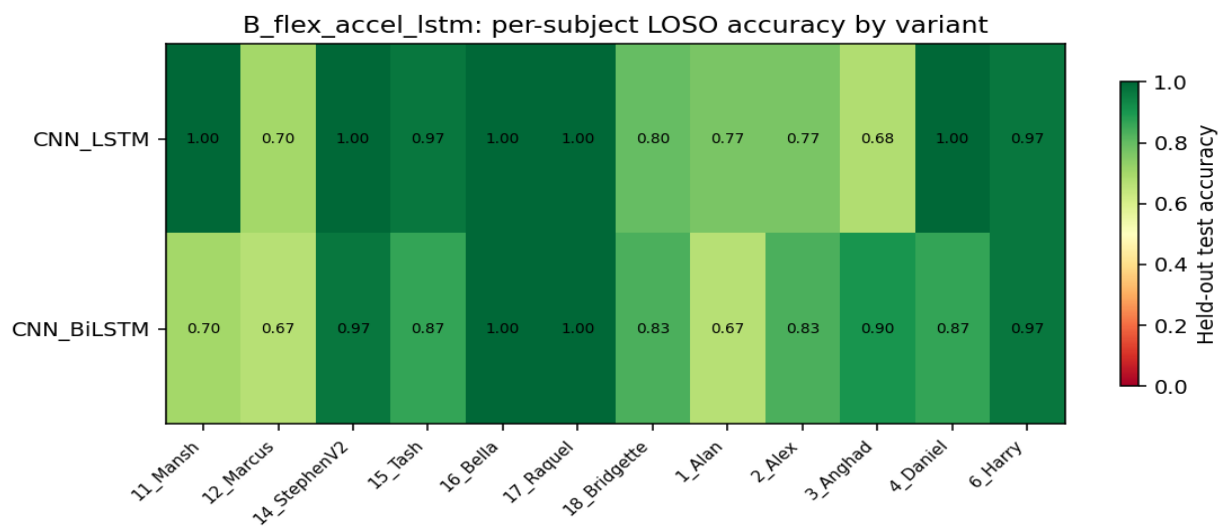
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_lstm	58,406	0.8870	0.1265	0.6774	1.0000	0.8730
CNN_BiLSTM	95,526	0.8558	0.1174	0.6667	1.0000	0.9063

Best variant by LOSO mean: **CNN_lstm**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	1.000	0.700
12_Marcus	331	0.700	0.667
14_StephenV2	331	1.000	0.967
15_Tash	331	0.967	0.867
16_Bella	331	1.000	1.000
17_Raquel	331	1.000	1.000
18_Bridgette	331	0.800	0.833
1_Alan	331	0.767	0.667
2_Alex	331	0.767	0.833
3_Anghad	330	0.677	0.903
4_Daniel	331	1.000	0.867
6_Harry	331	0.967	0.967

Variant: CNN_LSTM

Params: 58,406

LOSO mean accuracy: 0.8870 ± 0.1265 (min 0.6774, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	0/5	5/5	5/5	5/5	2/5	4/5	0.700
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
15_Tash	5/5	5/5	5/5	5/5	5/5	4/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
18_Bridgette	0/5	5/5	4/5	5/5	5/5	5/5	0.800
1_Alan	5/5	4/5	0/5	5/5	5/5	4/5	0.767
2_Alex	5/5	4/5	4/5	5/5	5/5	0/5	0.767
3_Anghad	5/5	5/5	5/5	0/5	1/6	5/5	0.677
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	4/5	5/5	0.967
MEAN	0.833	0.967	0.883	0.917	0.864	0.867	0.887

Variant: CNN_BiLSTM

Params: 95,526

LOSO mean accuracy: 0.8558 ± 0.1174 (min 0.6667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	5/5	5/5	5/5	1/5	0.700
12_Marcus	1/5	5/5	5/5	0/5	5/5	4/5	0.667
14_StephenV2	5/5	5/5	4/5	5/5	5/5	5/5	0.967
15_Tash	5/5	5/5	5/5	5/5	5/5	1/5	0.867
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
18_Bridgette	2/5	5/5	4/5	5/5	5/5	4/5	0.833
1_Alan	5/5	5/5	5/5	0/5	0/5	5/5	0.667
2_Alex	5/5	4/5	5/5	5/5	5/5	1/5	0.833
3_Anghad	5/5	5/5	5/5	5/5	3/6	5/5	0.903
4_Daniel	5/5	5/5	5/5	1/5	5/5	5/5	0.867
6_Harry	5/5	5/5	5/5	5/5	4/5	5/5	0.967
MEAN	0.883	0.900	0.967	0.767	0.858	0.767	0.856